

Notes: Gennaioli, Shleifer & Vishny (JFE, 2012)
Neglected Risks, Financial Innovation, and Financial Fragility

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1 Big picture

- **Question.** How can financial innovation create fragile securities and market crashes even without leverage, fire-sale constraints, or irrational ex post panic?
- **Core answer.** Financial intermediaries engineer securities that investors perceive as safe because they neglect unlikely but important risks. Since the neglected risks are not priced, issuance is excessive. When news makes the risk salient, investors realize that the new claims are false substitutes for safe traditional claims and flee back to safety.
- **Central mechanism.** A demand for safe cash flows plus local thinking generates over-issuance of securities that appear safe but are exposed to ignored bad states.
- **Key contrast.** Under rational expectations, innovation can create new claims, but investors price risk correctly and issuance is disciplined. Under local thinking, the same engineering technology generates too many new claims.
- **Fragility result.** Once a bad signal makes the neglected risk salient, the new claim's price can fall sharply while the traditional safe bond price rises. This generates flight-to-safety dynamics.
- **Production-economy result.** Innovation can distort real investment. Intermediaries may overinvest in the risky asset used to manufacture apparently safe claims. Once risk is recognized, both investors and intermediaries can be worse off.
- **Policy message.** Financial innovation is not always destabilizing, but when innovation exploits neglected risks and creates perceived safe assets, the private volume of issuance can be socially excessive.

2 Incremental contribution of the paper

- **Unifies innovation and fragility.** The paper connects the demand for safe assets, financial engineering, over-issuance, surprise risk realization, and flight to safety in one model.
- **Adds belief formation to financial innovation models.** Classic innovation models, such as Allen and Gale-style models, explain how intermediaries repackage cash flows. Gennaioli, Shleifer, and Vishny add local thinking: investors and possibly intermediaries fail to represent rare bad states.
- **Explains false substitutes.** The model formalizes why newly engineered securities can be treated as substitutes for safe assets and then suddenly cease to be substitutes once neglected risks become salient.
- **Fragility without leverage.** Standard crisis stories often rely on leverage, margin calls, or fire sales. This paper shows that large price movements and financial fragility can arise from belief revision and excessive supply even without those mechanisms.
- **Separates rational innovation from over-innovation.** Rational expectations can generate useful innovation and segmentation, but not the same over-issuance and fragility. The non-rational representation of risk is the key distortion.
- **Links macro safe-asset demand to intermediary behavior.** Investors' preferred habitat for safe cash flows induces intermediaries to manufacture claims that satisfy perceived safety demands.
- **Welfare contribution.** The paper shows that innovation can hurt investors in an exchange economy and hurt both investors and intermediaries in a production economy when innovation distorts real investment.

Interpretation: The incremental contribution is to show that financial innovation is not only a response to missing markets. It can also create fragility when the new securities exploit neglected risks and are therefore overproduced relative to their true risk.

3 Model primitives and measurement

This is a theoretical paper. The “data” in the model are states of the world, cash flows, signals, and beliefs. The empirical motivation comes from examples such as mortgage-backed securities, collateralized mortgage obligations, and money market funds, but the formal model is analytic rather than estimated.

3.1 Assets and states

There are three dates $t = 0, 1, 2$ and two underlying assets:

- Asset B is safe and pays $R > 1$ at $t = 2$.
- Asset A is risky and pays y_i in state $i \in \{g, d, r\}$, where g is growth, d is downturn, and r is recession.

The payoff ranking and probabilities satisfy

$$y_g > 1 > y_d > y_r, \quad \pi_g > \pi_d > \pi_r. \quad (1)$$

Interpretation: Asset B is the benchmark safe claim. Asset A is the risky pool from which intermediaries can engineer new claims that appear safe.

3.2 Investors and preferred habitat for safety

Investors have utility

$$U = E[C_0 + C_1 + \theta \min(C_{2g}, C_{2d}, C_{2r})], \quad \theta > 1. \quad (2)$$

They value consumption in the worst state and therefore have extreme demand for safe $t = 2$ cash flows.

Interpretation: This is a preferred-habitat formulation. Investors want cash flows that are safe across all states. The model uses extreme risk aversion for clarity, but the economic force is strong demand for safe, information-insensitive claims.

3.3 Signals and belief revision

At $t = 1$, agents observe a signal $s \in \{\bar{s}, \underline{s}\}$ about the payoff of the risky asset. The bad signal makes recession more salient. The signal is especially informative about the recession state.

Interpretation: The signal is not just news about expected payoffs. It changes which states agents represent when evaluating securities. Under local thinking, the recession state may be neglected at $t = 0$ and represented after bad news at $t = 1$.

3.4 Traditional and innovative claims

Traditional securities are:

- a safe bond on asset B that pays R ;
- a risky share in asset A .

Financial innovation creates a new claim out of asset A that promises to pay R in all states. The maximum riskless volume under rational expectations is

$$f^{RE} = \frac{y_r}{R}. \quad (3)$$

Under local thinking, when the recession state is ignored, the apparent safe issuance capacity is instead

$$f^L = \frac{y_d}{R} > f^{RE}. \quad (4)$$

Interpretation: This inequality is the engine of over-issuance. If investors ignore the recession state, intermediaries can issue claims that look safe through the downturn state but are not actually safe in the recession state.

4 Models and theoretical specifications

4.1 Rational expectations with traditional claims

With only traditional claims, the equilibrium has all safe bonds bought by investors and risky shares retained by intermediaries:

$$a = 0, \quad b = 1, \quad p_A = E[y], \quad p_B = \theta R. \quad (5)$$

Interpretation: Investors absorb the safe bond because they value safety. The risky asset remains with the intermediary because investors value it at its worst payoff while the intermediary values it by expected payoff.

4.2 Rational expectations with innovation

With innovation under rational expectations, the intermediary issues the maximum truly safe volume:

$$f^{RE} = \frac{y_r}{R}. \quad (6)$$

The new claim sells at the same price as the traditional bond:

$$p_N = p_B = \theta R. \quad (7)$$

Interpretation: Rational innovation relaxes the shortage of safe claims, but it does not generate false safety. Investors understand the recession state and only buy claims that are truly safe in that state.

4.3 Local thinking with innovation at $t = 0$

Under local thinking, investors and intermediaries initially neglect the recession state. The new claim is treated as safe if it pays in the growth and downturn states. Thus the issuance volume becomes

$$f^L = \frac{y_d}{R}. \quad (8)$$

Since $y_d > y_r$, issuance is excessive relative to the rational benchmark.

Interpretation: The new claim is a false substitute. It mimics the safe asset in represented states, but it is exposed to the neglected recession state.

4.4 Local thinking and news at $t = 1$

After a bad signal, investors recognize that the recession state is possible. They revalue the new claim as risky. The price of the traditional bond rises toward its reservation value, while the price of the new claim drops.

$$\text{bad news} \Rightarrow p_B \uparrow, \quad p_N \downarrow. \quad (9)$$

Interpretation: The market segmentation between old safe claims and new risky claims is endogenous. The same cash-flow package that looked like a safe substitute at $t = 0$ becomes a risky false substitute at $t = 1$.

4.5 Production economy

The paper then lets intermediaries choose real investment in A and B . When safe claims are in short supply, local thinking creates strong incentives to invest in the risky asset A because it can be used to manufacture apparently safe new claims.

Interpretation: Innovation now distorts real investment, not only financial claims. The safe-asset demand pushes resources toward the risky technology used to create false substitutes.

4.6 Welfare

In the exchange economy, innovation can benefit intermediaries while harming investors. In the production economy, excessive issuance and distorted investment can make both investors and intermediaries worse off.

Interpretation: The welfare result is subtle. Innovation expands trade, which is normally beneficial, but if it expands trade in false substitutes based on neglected risks, the resulting allocation can reduce welfare.

5 Identification and theoretical design

This is not an empirical identification paper. The relevant design is theoretical: the authors compare rational expectations and local thinking under the same economic environment.

5.1 Main theoretical comparison

The model holds fixed:

- the supply of safe and risky cash flows;
- investor demand for safety;
- intermediary ability to repackage cash flows;
- the signal structure.

It changes the representation of rare risks.

Rational expectations: all states represented, (10)

Local thinking: unlikely bad states neglected at $t = 0$. (11)

Interpretation: This comparison identifies the effect of neglected risk. The difference between f^{RE} and f^L is not caused by technology or preferences, but by belief representation.

5.2 Why local thinking is needed

Without neglected risk, innovation may create useful safe cash flows but does not create false substitutes. The rational benchmark can generate new issuance, but not the same over-issuance and fragility.

Interpretation: The paper's central identification claim is conceptual: the false-substitute phenomenon requires a deviation from rational representation of low-probability states.

5.3 Role of intermediary wealth

After the bad signal, intermediaries may want to buy back the new claims, but their wealth can be limited. If their wealth is small relative to the volume of false substitutes, prices of new claims collapse.

Interpretation: Intermediary balance-sheet capacity matters even without leverage. The intermediaries' ability to absorb newly recognized risk determines how large the price drop becomes.

5.4 Rational intermediaries with local-thinking investors

The paper also considers rational intermediaries selling to locally thinking investors. The main fragility mechanism survives: intermediaries can profit by issuing claims that investors wrongly view as safe.

Interpretation: The model does not require all agents to be confused. It is enough that investors demand false substitutes and intermediaries have incentives to supply them.

6 Tables and figures

6.1 Figure 1: Timing of the model; Figure 2: Equilibrium without innovation

Read:

- Figure 1 lays out the three-date timing: financial claims are traded at $t = 0$, signals and re-trading occur at $t = 1$, and payoffs are realized at $t = 2$.
- Figure 2 shows the rational-expectations equilibrium without innovation. Investors buy the safe bond, intermediaries keep risky shares, and bond prices reflect investor demand for safety.

Economic message: Figure 1 defines the dynamic structure. Figure 2 is the benchmark shortage-of-safety environment before innovation. These figures make clear that fragility is not driven by uncertainty about timing or trading opportunities. The key change later is how new claims are priced and repriced when risk representation changes.

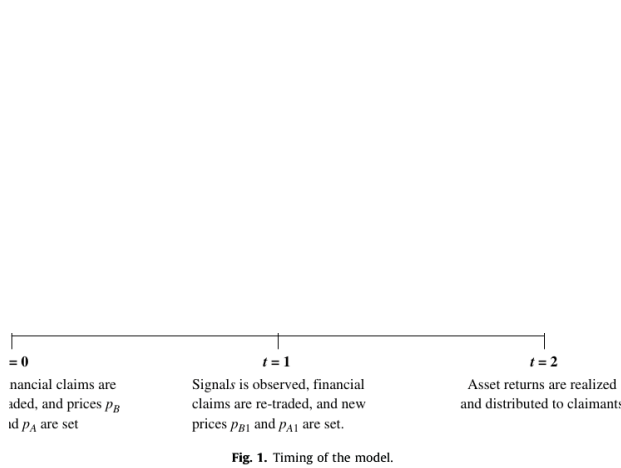


Figure 1: Timing of the model. The diagram separates initial issuance, signal revelation, and final payoffs.

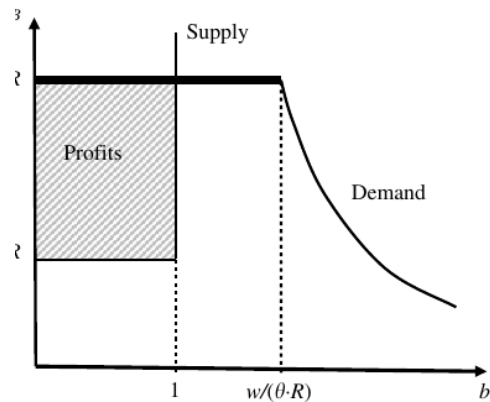


Figure 2: Equilibrium without innovation. Safe bonds are scarce and valuable to investors.

6.2 Figure 3: Equilibrium with innovation under rational expectations; Figure 4: Equilibrium with financial innovation under local thinking

Read:

- Figure 3 shows rational innovation: new claims expand the supply of safe securities, but only up to the truly safe issuance volume.
- Figure 4 shows local-thinking innovation: neglect of the recession state shifts the perceived supply of safe claims too far to the right.
- In Figure 4, the new supply can push prices below the no-innovation level or create a large issuance glut.

Economic message: This is the core visual comparison. Rational innovation fills a safe-asset shortage; local-thinking innovation overfills it by manufacturing claims that look safe but are not. The difference between the diagrams is not the engineering technology. It is the set of states investors consider. That is why the same financial engineering technology can be socially useful under rational expectations and fragile under neglected risk.

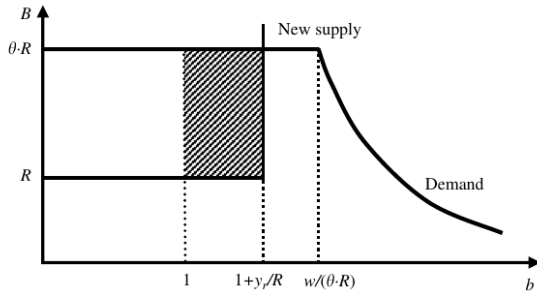


Fig. 3. Equilibrium with innovation under rational expectations.

Figure 3: Rational innovation. New claims are issued only up to the truly safe level.

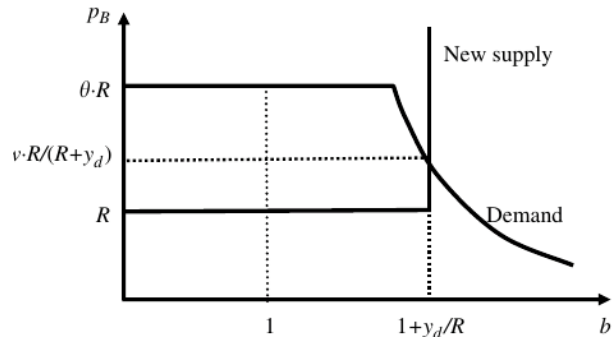


Fig. 4. Equilibrium with financial innovation under local thinking.

Proposition 1. Under local thinking, the volume of innovation is too large.

Figure 4: Local-thinking innovation at $t = 0$. Neglected recession risk permits excess issuance.

6.3 Figure 5: Equilibrium at $t = 1$ with financial innovation under local thinking; Figure 6: Financial innovation and market volatility

Read:

- Figure 5 shows the market after bad news makes the neglected recession state salient. The old bond market and the new-claim market separate.
- The traditional bond price rises because investors fly to truly safe claims.
- The new claim price falls because investors no longer regard it as a safe substitute.
- Figure 6 summarizes volatility: with innovation, good-state valuations rise but bad-news valuations fall much more sharply.

Economic message: These figures show the fragility mechanism. The crisis occurs because over-issued new claims are revealed to be false substitutes for safety. Figure 6 is the paper's main qualitative output. Innovation creates a boom in good times because investors think they have more safe assets. It creates a bust after bad news because the same claims no longer provide safety.

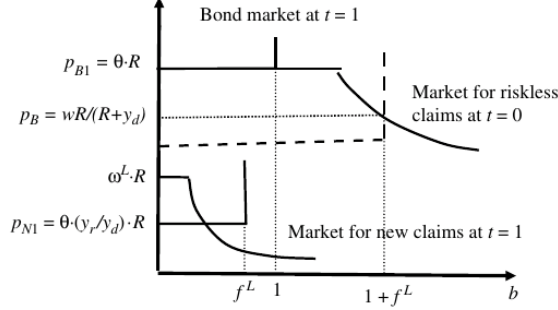


Fig. 5. Equilibrium at $t=1$ with financial innovation under local thinking.

Figure 5: Local-thinking innovation at $t = 1$. Old and new claims separate after risk becomes salient.

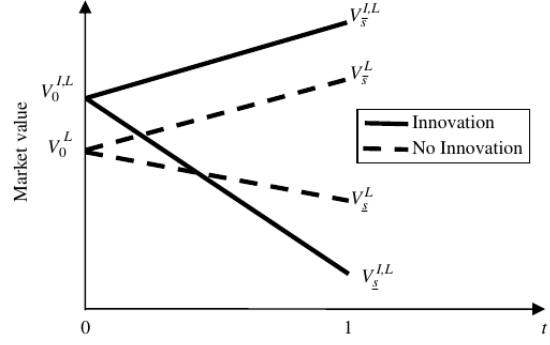


Fig. 6. Financial innovation and market volatility.

Figure 6: Innovation and market volatility. Innovation amplifies good-state gains and bad-state losses.

6.4 Notes-created table: model results and interpretation

Result	Formal content	Interpretation
Lemma 1	With traditional claims under rational expectations, investors buy the safe bond and intermediaries retain risky shares.	Preferred habitat for safety creates a shortage of safe claims.
Lemma 2	Rational innovation issues $f^{RE} = y_r/R$ safe claims.	Innovation is useful but disciplined by the true worst state.
Proposition 1	Local thinking permits issuance $f^L = y_d/R > f^{RE}$.	Neglected recession risk creates over-issuance.
Proposition 2	After bad news, old safe claims rise in value and new claims fall.	False substitutes are revealed and markets become fragile.
Corollary 1	Innovation amplifies market volatility under local thinking.	The boom-bust pattern is a feature of belief revision and over-supply.
Proposition 3	In production, local thinking can distort real investment toward risky assets.	Financial innovation can change real allocations, not just prices.
Proposition 5	When innovation distorts investment, everyone can be worse off.	Innovation can be privately profitable and socially harmful.

Notes-created table: Main model results. This table is not an original table from the paper. It condenses the lemmas and propositions into the mechanism connecting neglected risk, innovation, over-issuance, and fragility.

7 Short summary

- Investors demand safe cash flows.
- Intermediaries can repackage risky cash flows into new claims that appear safe.
- Under rational expectations, new claims are issued only to the truly safe level.
- Under local thinking, investors neglect unlikely recession risk and treat risky claims as safe.
- This creates excessive issuance of false substitutes.
- When news makes neglected risk salient, the new claims are repriced downward.
- Investors fly back to traditional safe claims, raising their price.
- Intermediary wealth limits can deepen the crash even without leverage.
- In a production economy, false substitutes can distort real investment.
- Innovation can therefore reduce welfare when it exploits neglected risks.

8 Conclusion

8.1 Core conclusion

Gennaioli, Shleifer, and Vishny show that financial innovation can create fragility when intermediaries manufacture securities that investors perceive as safe because they neglect unlikely risks. The problem is not innovation per se, but innovation that transforms risky cash flows into apparently safe claims under a distorted representation of the state space.

8.2 Economic intuition

The intuition is straightforward. Investors want safe claims. Intermediaries can carve apparently safe claims from risky assets. If everyone ignores the recession state, the new claims look safe and are issued in large quantities. When a bad signal makes the recession state salient, investors realize that the new claims are not safe. The old safe asset becomes more valuable, while the new claims collapse in value.

8.3 Incremental contribution revisited

The paper's incremental contribution is to combine preferred habitat for safe assets, financial engineering, and local thinking into one mechanism. This mechanism explains why innovation can simultaneously expand credit in booms and create fragility in busts. It also explains why rational expectations models of innovation miss the central crisis feature: false substitutes that turn out not to be safe.

8.4 Policy implications

The model suggests that regulators should focus not only on leverage and capital requirements but also on the structure of newly manufactured safe assets. If innovation creates securities whose safety depends on ignoring rare states, then disclosure, stress testing, capital charges, and limits on ratings-based demand may reduce excessive issuance.

8.5 Limitations

The model is stylized. It uses extreme risk aversion, simplified states, and a reduced-form representation of local thinking. It does not estimate the size of neglected-risk distortions. It also abstracts from many institutional details of securitization, ratings, collateral, liquidity regulation, and banking leverage.

8.6 Takeaway

The main takeaway is that financial innovation can be fragile when it supplies what investors want by hiding what they neglect. A security that looks safe in represented states can become dangerous once neglected states become salient. This is why innovation can create both a boom in perceived safety and a bust when the false substitute is revealed.